Lab Sheet 3: SQL Functions

A. Number Functions

Demonstrate ABS() with suitable examples.

```
89      #sheet3
90 •    SELECT ABS(-25) AS Absolute_Value;
```

Result Grid | Filter Rows: | Expo

Absolute_Value
25

1. Demonstrate ROUND() with suitable examples.

```
91 •    SELECT ROUND(125.678, 2) AS Rounded_Value;
```

Result Grid | Filter Rows: | Export:

Rounded_Value
125.68

2. Demonstrate CEIL() and FLOOR().

```
92 •    SELECT CEIL(12.34) AS Ceiling_Value,
93             FLOOR(12.78) AS Floor_Value;
```

Result Grid | Filter Rows: | Export:

Ceiling_Value	Floor_Value
13	12

3. Demonstrate MOD() with suitable examples.

```
94 •    SELECT MOD(17, 5) AS Remainder;
95
```

esult Grid | Filter Rows: | Expo

Remainder
2

4. Calculate the rounded salary of employees.

```
95 •    SELECT Name, Salary, ROUND(Salary, -2) AS Rounded_Salary FROM Employee;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name	Salary	Rounded_Salary
Amit	42350.00	42400
Rahul	50820.00	50800
Neha	60500.00	60500
Priya	45980.00	46000
Rohit	33880.00	33900

B. Aggregate Functions

6. Find the total number of students using COUNT().

```
96 •    SELECT COUNT(*) AS Total_Students FROM Student;
```

esult Grid | Filter Rows: | Export: | Wrap

Total_Students
14

7. Find the total salary of all employees using SUM().

```
97 •    SELECT SUM(Salary) AS Total_Salary FROM Employee;
```

Result Grid | Filter Rows: | Export: | Wrap Cell C

Total_Salary
233530.00

8. Find the average salary using AVG().

```
98 •  SELECT AVG(Salary) AS Average_Salary FROM Employee;
99
```

Result Grid | Filter Rows: | Export: | Wrap Cell Conte

Average_Salary
46706.000000

9. Find the highest salary using MAX().

```
99 •  SELECT MAX(Salary) AS Highest_Salary FROM Employee;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Co

Highest_Salary
60500.00

10. Find the lowest salary using MIN().

```
100 •  SELECT MIN(Salary) AS Lowest_Salary FROM Employee;
101
```

Result Grid | Filter Rows: | Export: | Wrap Cell Con

Lowest_Salary
33880.00

C. Character Functions

11. Demonstrate UPPER() on student names.

```
101 •  SELECT Name, UPPER(Name) AS Upper_Name FROM Student;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Co

Name	Upper_Name
Anshika	ANSHIKA
Radha	RADHA
Jayshree	JAYSHREE
Ruhani	RUHANI
Shreya	SHREYA
Maya	MAYA
Rahul	RAHUL
Priya	PRIYA
Aman	AMAN
Neha	NEHA
Rohit	ROHIT
Simran	SIMRAN

12. Demonstrate LOWER() on student names.

```
102 •  SELECT Name, LOWER(Name) AS Lower_Name FROM Student;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Co

Name	Lower_Name
Anshika	anshika
Radha	radha
Jayshree	jayshree
Ruhani	ruhani
Shreya	shreya
Maya	maya
Rahul	rahul
Priya	priya
Aman	aman
Neha	neha
Rohit	rohit

13. Demonstrate LENGTH() on student names.

```
103 •  SELECT Name, LENGTH(Name) AS Name_Length FROM Student;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Conter

Name	Name_Length
Anshika	7
Radha	5
Jayshree	8
Ruhani	6
Shreya	6
Maya	4
Rahul	5
Priya	5
Aman	4
Neha	4
Rohit	5
Simran	6

14. Demonstrate CONCAT() using first and last names.

15. Demonstrate SUBSTRING()/SUBSTR() with suitable examples.

```
104 •  SELECT First_Name, Last_Name, CONCAT(First_Name, ' ', Last_Name) AS Full_Name FROM Student;
105 •  SELECT Name,SUBSTRING(Name, 1, 3) AS First_Three_Characters FROM Student;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name	First_Three_Characters
Anshika	Ans
Radha	Rad
Jayshree	Jay
Ruhani	Ruh
Shreya	Shr
Maya	May
Rahul	Rah
Priya	Pri
Aman	Ama
Neha	Neh
Rohit	Roh
Simran	Sim

D. Conversion Functions

16. Convert a number into a character/string value.

```
106 •  SELECT CAST(12345 AS CHAR) AS String_Value;
```

Result Grid | Filter Rows: | Export: | Wrap C

String_Value
12345

17. Convert a string into a numeric value.

```
107 •  SELECT CAST('12345' AS UNSIGNED) AS Numeric_Value;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Co

Numeric_Value
12345

18. Convert a date value into a different format.

```
108 •  SELECT DATE_FORMAT('2026-09-17', '%d-%m-%Y') AS Formatted_Date;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Formatted_Date
17-09-2026

19. Demonstrate CAST() with suitable examples.

```
109 •  SELECT CAST(125.75 AS DECIMAL(10,2)) AS Converted_Value;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Converted_Value
125.75

20. Demonstrate CONVERT() where supported by the DBMS.

```
110 •  SELECT CONVERT(12345, CHAR) AS String_Value;
```

Result Grid | Filter Rows: | Export: | Wr

String_Value
12345

E. Date Functions

21. Display the current date using a suitable date function.

```
111 •  SELECT CURDATE();
```

Result Grid | Filter Rows:

CURDATE()
2026-09-17

22. Extract the year, month and day from a date.

```
112 •  SELECT YEAR('2026-09-17') AS Year,MONTH('2026-09-17') AS Month,DAY('2026-09-17') AS Day;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Year	Month	Day
2026	9	17

23. Find the difference between two dates.

```
113 •  SELECT DATEDIFF('2026-09-17', '2026-01-01') AS Date_Difference;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Date_Difference
259

24. Add a specified number of days to a date.

```
114 •  SELECT DATE_ADD('2026-09-17', INTERVAL 10 DAY) AS New_Date;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

New_Date
2026-09-27

25. Display employees' joining dates in a required date format.

```
15 •  SELECT Name, Joining_Date,DATE_FORMAT(Joining_Date, '%d-%m-%Y') AS Formatted_Joining_Date FROM Employe;
```